

Personal Identifiability of User Tracking Data During VR Training

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ABSTRACT

Recent research indicates that user tracking data from virtual reality (VR) experiences can be used to personally identify users at accuracies as high as 95 percent. However, these results indicating that non-verbal data should be understood as personally identifying data were based on observing 360-degree videos. In this paper, we present participant identification results based on a session of user tracking data from a VR training application, which show accuracies above 90 percent. While still highly accurate, this decrease indicates that the personal identifiability of user tracking data is likely dependent upon the nature of the underlying VR experience.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality; Computing methodologies—Machine learning—Cross-validation

1 INTRODUCTION

In recent work, Miller et al. [5] were able to identify participants (N=511) with accuracies up to 95% by training common machine learning (ML) models on motion data captured from participants observing 360° videos. In their paper, one described limitation is that the Virtual Reality (VR) experiences were primarily stationary with interactions largely limited to standing with little motion. Too much homogeneity in the data collection process could increase the identifiability of individual users as it can limit the realism of the collected data.

We implement the approach presented by Miller et al. [5] on data that we previously captured from a user study (N=60) to address this limitation. The VR experience in our user study involved training the participant how to troubleshoot a surgical robot, requiring lots of motion, unlike the primarily stationary data of Miller et al. [5].

In our analysis, we found that two of the three ML models performed noticeably worse than the results reported by Miller et al. [5]. This is surprising considering that our data set is much smaller, i.e., random selection has higher probability of identifying the correct participant (1/60) than the original study (1/511). We believe this indicates that the virtual experience affects personal identifiability.

2 RELATED WORKS

VR provides a device context that allows software to incorporate several pieces of behavioral biometric data that is normally inaccessible. Mustafa et al. [9] were able to use machine learning on HMD data to authenticate users with equal error rates as low as 7%. More recent works explore the use of biometric data not exclusively for authentication, but also for identification. Pfeuffer et al. [11] recently were able to identify participants at rates as high as 63%. Work by Miller et al. [5] has shown positive identification rates above 95% when identifying among 511 participants.

Other uses of the rich data afforded by VR data include predicting procedural knowledge gain [8] and cybersickness [2], indicating that the collection of this data will likely soon be more commonplace.

3 VR LEARNING STUDY

For this study, we used an existing VR application that was designed to train first assists how to troubleshoot a surgical robot in a robotic operating room (OR) [6]. This application makes use of virtual hand selections and manipulations [4, 7] to support multiple interactions with equipment, as well as selecting dialog options to communicate with a virtual surgeon and non-sterile staff member. In order to complete the scenario, the user must perform a variety of ordered steps involving these interactions, move about the virtual OR using real walking [3], and look at notifications and an endoscopic monitor displayed on a vision cart. Participants were guided by interaction cues [1] which conveyed which action to take.

3.1 Materials

This study used the HTC Vive system, which includes an HMD and two handheld controllers to interact with the VR training application. The Vive HMD has a 90Hz refresh rate, a display resolution of 1080x1200 pixels per eye, and affords a 110° diagonal field of view. The HMD was fitted with the Vive audio strap that integrates over-the-ear headphones. The VR application was developed in Unity and maintained 90 frames per second to match the Vive's refresh rate. The VR training application also collected HMD and controller tracking data every frame.

3.2 Procedure

The following procedure was reviewed and approved by the University Institutional Review Board (IRB). A total of 60 participants were recruited through university mailing lists. After informed consent, the session began with a background survey on the participant's demographics, education, and technology experience. The experimenter would then help the participant don the HTC Vive and run the SteamVR tutorial to train them how to use the Vive. The participant would then experience the VR training application. After completing the application, the participant was administered a number of questionnaires regarding their VR experience. Finally, participants were administered a knowledge test consisting of 20 multiple-choice questions pertaining to the training scenario. This session lasted approximately 60 minutes.

4 IDENTIFYING PARTICIPANTS

We followed the basic approach described by Miller et al. [5] in terms of feature representation and general ML approaches, including the use of the same 90-valued feature vector representation for each second. These sets of feature vectors were then partitioned into 10 subsessions for each participant. In keeping with Miller et al. [5], we trained three different ML models: k-Nearest-Neighbors (kNN), random forests (RF), and gradient boosting machines (GBM). RF and GBM were implemented with scikit-learn defaults [10]. After being trained, all three of these classifiers attempt to solve our multiclass classification problem – given a single feature vector, predict which participant generated that feature vector.

We applied 20-fold leave-one-out cross-validation to compare the performance of the three approaches, retaining one subsession per

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participant for evaluating the performance of the classifier trained on the other 9 subsessions. We used the plurality of these predictions at the subsession level, saying each classifier ultimately identified the most commonly predicted identity for that subsession. Each training and validation process yields a percentage of correct identifications out of all 60 participants, and we averaged this value over all 20 cross-validations to get a mean accuracy rating for each classifier. Our results are shown in Figure 1. Our GBM accuracy was higher than that reported by Miller et al., however our kNN and RF classifiers performed worse than those of Miller et al. [5].

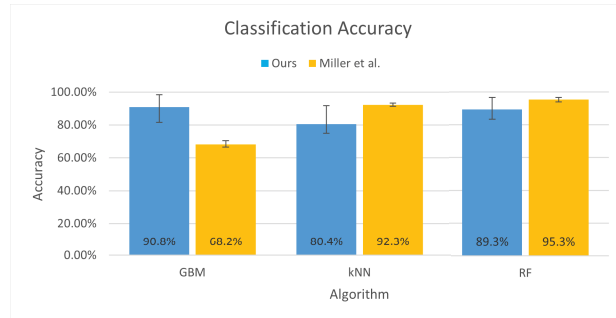


Figure 1: Classification accuracy averaged over 20 Monte-Carlo cross-validations. Error bars indicate range of values. Error bar values for Miller et al. estimated from Figure 3 of Miller et al. [5].

5 RESULTS

We showed that our classifiers were able to achieve fairly accurate identification of users, much higher than chance (1/60), but given that there are only 60 potential classifications, our result of 90.83% seems fairly weak in comparison to recent work by Miller et al. [5], who were able to achieve greater than 95% identification accuracy among 511 participants. One explanation for this disparity is the large difference in VR experiences presented to the users. While our experiment made use of a VR training scenario with several virtual object manipulations that required movement around the virtual space, the environment presented in the Miller et. al [5] consisted of five 360° videos and five in-VR questionnaires, presented one after each video. Indeed, the authors stated that “a limitation of this work is that both tasks captured standing users with little motion ... the generalizability beyond the tasks demonstrated here should be tested.” [5]. Our results seem to indicate that while it’s still quite possible to identify a user in a scenario with more movement, the classifiers seem to perform differently.

6 LIMITATIONS

First, in both this work and that of Miller et al. [5], cross-validation should have been used to determine the hyperparameters of the different models (e.g., the k in kNN, the number and depth of trees in the RF algorithm). Note that in both this work and in Miller et al. [5], the ML models are quite limited in that they do not take advantage of the fact that this data is actually temporal. Similarly, in this work, we divided the motion data of each participant into ten equal size pieces—a somewhat arbitrary slicing of their motion data.

Second, we note that, if our aim were to actually identify participants from motion data, we would use cross-validation to select the best model/hyperparameters on, say, the learning data set, train a model on the full learning data set using the selected model/hyperparameters, and then apply this model to a new unseen data set. That is, care should be taken not to use performance on the test set to draw conclusions about the best model to pick.

7 CONCLUSION

In this paper, we investigated the feasibility of implementing the approach presented by Miller et al. [5] to identify users in a VR scenario that involved more than stationary users and little motion. We ultimately determined that while these classifiers performed better than random chance at correctly identifying participants, their performance is worse than the prior work’s results. While there are several factors contributing to this discrepancy, it seems as though the homogeneity in their data set led to easier identification of participants. Further work is needed to establish reasonable expectations of personal identifiability of user tracking data in VR.

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